Supplementary material

Why genomic prediction of colistin resistance does not transfer between *Klebsiella pneumoniae* collections

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Supplementary tables

Supplementary Table S1. Colistin sensitivity and specificity for each rule, with and without the targeted *mgrB* disruption check, on broth-microdilution labels (n = 1,528; 367 resistant). Plotted in Figure 1.

Rule	Sensitivity (95% CI)	Specificity (95% CI)
AMRFinderPlus	28.3% (24.0–33.2)	97.8% (96.7–98.5)
AMRFinderPlus + *mgrB* disrupted	**56.7% (51.6–61.7)**	**95.2% (93.8–96.3)**
Kleborate	39.5% (34.6–44.6)	96.3% (95.0–97.2)
Kleborate + *mgrB* disrupted	**45.5% (40.5–50.6)**	**95.3% (94.0–96.4)**
mgrB disrupted alone	41.7% (36.8–46.8)	96.5% (95.2–97.4)

Supplementary Table S2. Sensitivity of the *mgrB* disruption rule in each held-out study (leave-one-study-out). Plotted in Figure 2.

Held-out study	Resistant	Sensitivity, disruption only
PRJEB10018	162	39.5%
PRJNA504600	27	48.1%
PRJEB22890	63	50.8%
PRJNA288601	20	55.0%
PRJNA292904	7	57.1%
PRJNA307517	19	63.2%
PRJEB10561	17	70.6%
Pooled	**315**	**47.0% (41.5–52.5)**

Supplementary Table S3. Balanced accuracy of the per-drug rule, logistic regression and gradient boosting under each split scheme. Plotted in Figure 3.

Split	Rule + *mgrB*	Logistic regression	Gradient boosting
Random	0.759	**0.844**	**0.842**
Study-aware	0.759	**0.822**	**0.810**
Lineage-aware	**0.759**	0.731	0.709

Supplementary Table S4. Positive and negative predictive value of the AMRFinderPlus plus *mgrB* disruption rule (sensitivity 56.7%, specificity 95.2%) across plausible colistin resistance prevalences. Sensitivity and specificity are properties of the rule; predictive values also depend on how common resistance is in the population tested. Prevalences are illustrative and are not estimates for any particular setting.

Colistin resistance prevalence	PPV	NPV
2%	19.4%	99.1%
5%	38.3%	97.7%
10%	56.8%	95.2%
24% (this cohort)	78.9%	87.4%